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Conference 2000



Security Beyond the Enterprise

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Security in Layers

- Physical, you have to pry the device from their cold dead fingers
- Access control, is there a password when you turn it on?
- Authorization, App level passwords
- Auditing, can you trace useage/abuseage?
- Encryption, is the data protected?
- Wireless encryption, is it safe to broadcast?
- Are actions non-repudiable?

A Lot Can Happen in a Year!

- More physical security devices
- More sophisticated access control products
- Improved OS level security
- Improved security application
- First PKI and certificate-based products
- First biometric devices
- First Viruses
- More developers, Tools and more opportunity!

New Software Products and Revs

- OnlyMe 2.08
- Tealock 3.01
- Commander 1.45
- Sign-On 2.0
- AmFound 1.2
- ID Palm 1.16
- SafetyPIN 1.5
- Restrictor
- Enforcer
- pAutolock1.2
- PadlockHack1.5
- PGP for Palm
- EasyLock 2.2
- JotLoc 1.5
- Padlock plus 1.1
- Autolock 3.6
- Matrix

Other Categories…

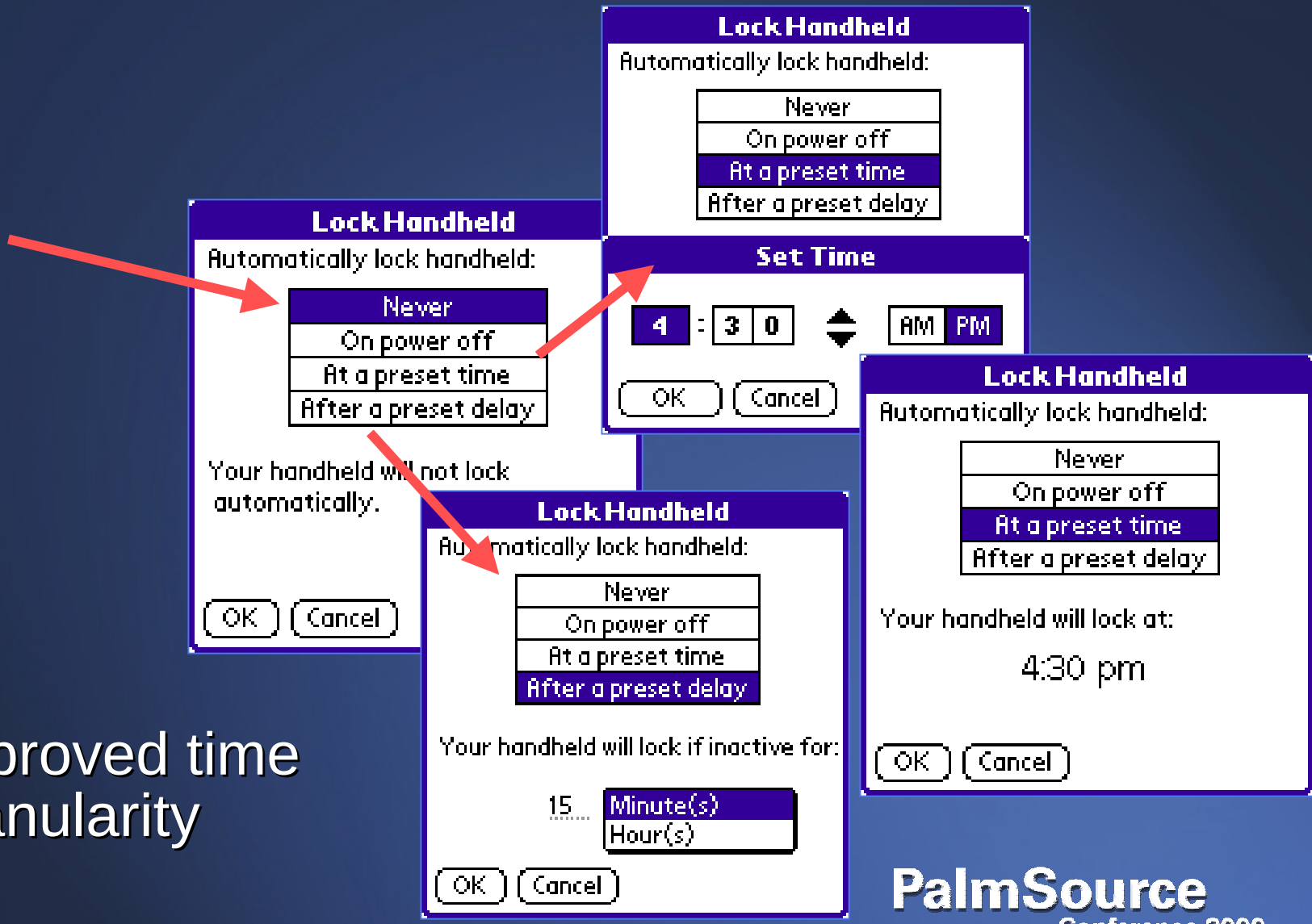
- Password lists
- Password generators
- Account information holders
- Encryption for Memos, addresses, etc
- If-found-then-call programs
- Various off-and-lock hacks
- Over 100 security related products in all

Improved OS level security

- Small improvements in 4.0

- Developer shortcuts 1, 2, 5 will not work if a password is set.
- all shortcuts work normally if a password is NOT set
- Hitting a PC that is waiting for HotSync® packets wont crash the machine
- Bigger plans for the future

Improved Security Application 4.0



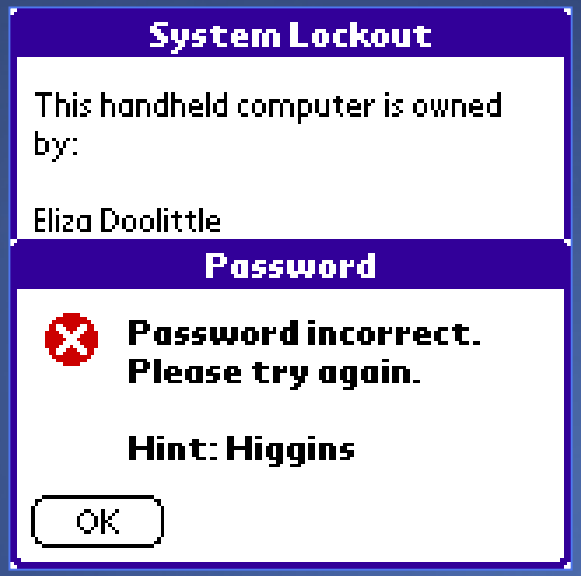
- Improved time granularity

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Improved Security Application

- Password hinting for the seriously memory challenged





Third Party Developments

- PKI
- VPN
- Biometrics
- Anti-Virus

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PKI and Certificate-Based Products

- Products from Verisign, Certicom/Mobiltrust, and others
- Complex systems for users to understand
- Useful for user authentication
- Issues include:-
  - Certificate standardisation
  - Acceptance of PKI systems in large deployments

First VPN Clients

- Beta products from Certicom being tested today, works with VPN gateways from Checkpoint, Cisco and Nortel
- V-One smaller client requires proprietary server behind the Firewall
- Not an easy task!

First Biometric Devices

- Applied Biometrics, Mytech
- Fingerprints
- Issues:-
  - Power
  - Battery life
  - Processing
  - Ease of use
- Alternates
  - Voice, body EMF, signature



First Viruses

- ….or at least attempts!!

- Bandwagon effect everyone has a cure!
- No real problematic viruses
- “accidental” deletion of code
- We DO take the threat seriously
- Working with Symantec, Trend, F-Secure, and CA

Opportunities

- SD cards, protected content
- User authentication
- Coping with passwords and ID’s
- Multi-use devices, personal and corporate data
- Beaming credentials
- Physical locks/leashes/carrying cases
- E-wallets and Net identity/personalization
- Better UI design than defaults
- Smart card readers and infrastructure
- Smart card emulation

Enterprise Issues and Opportunities

- Feedback from Palm Advisory council
  - Security, security, security
  - Encrypt addresses, calendars, notes, todo's etc.
  - Opportunities to develop enterprise level Apps.
  - Security Policy adherence
  - The Personal/Corporate dilema
  - Pervasive SSL
  - Central device management and secure software deployment

Guest Speakers

- Tom Traeger—Physical security
- Alex Robinson—Access control
- Cory Krutis—Multi-user access
- Steve Samson—New Security Toolkit

Physical Security
eHolster, Tom Traeger

- This evening Tom is modeling the eHolster
- Combines phone, devices, keyboards etc.
- The first real digital tool belt
- Attractive yet hormonal
- Modular



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Access Control: OnlyMe, Alex Robinson

- Access Security—What is important?
  - 1) Technical security
  - 2) Automatic operation
  - 3) Fast, easy password entry
  - 4) Seamless operation

1) Technical Security

- Don't allow brute-force hacking
- Don't allow communications
- Don't show the password on-screen
- Lock inside programs that don't allow exiting

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2) Automatic Operation

- The device locks without user intervention
- No need to think about locking the device

3) Fast, Easy Password Entry

- Gesture
- Applications buttons
- Graffiti
- On-screen buttons

4) Seamless Operation

- After unlocking, device is in exactly the state it was in before locking

Enterprise Access Controls Restrictor, Cory Krutis

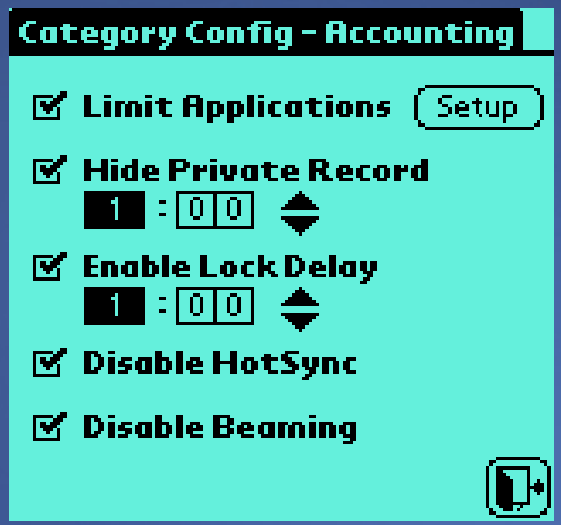


- Based upon Administrator / User Profile
- Allows the Administrator:
  - To decide what applications the user can have on their device
  - What data can be accessed during a HotSync
  - Turn the IR port on or off
  - Require the user to enter the password
  - Create a time delay for password entry—delay device locking

Enterprise Access Controls Restrictor, Cory Krutis



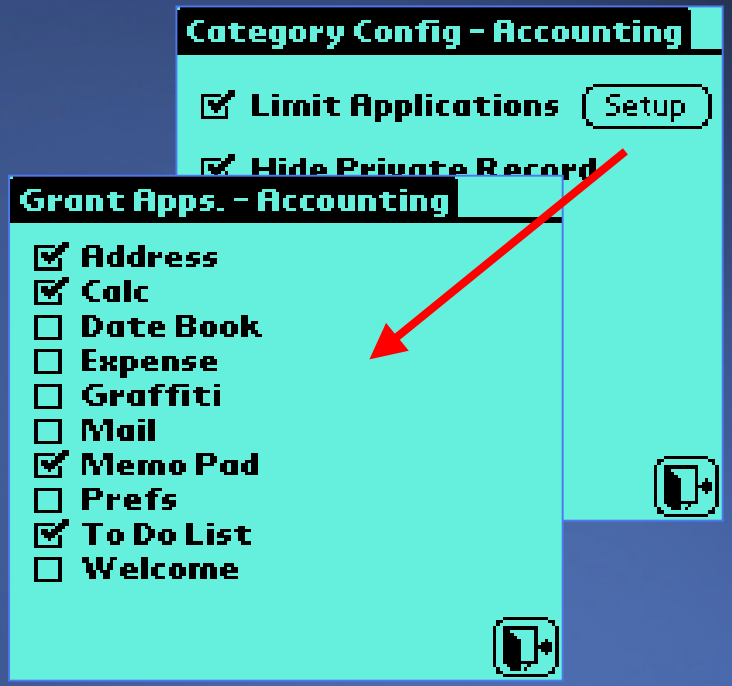
- Configuration allows you to:
  - Limit application access
  - Time delay for hiding private records
  - Time delay for locking down your device
  - Disable/Enable HotSync
  - Disable/Enable IR Port



Enterprise Access Controls Restrictor, Cory Krutis



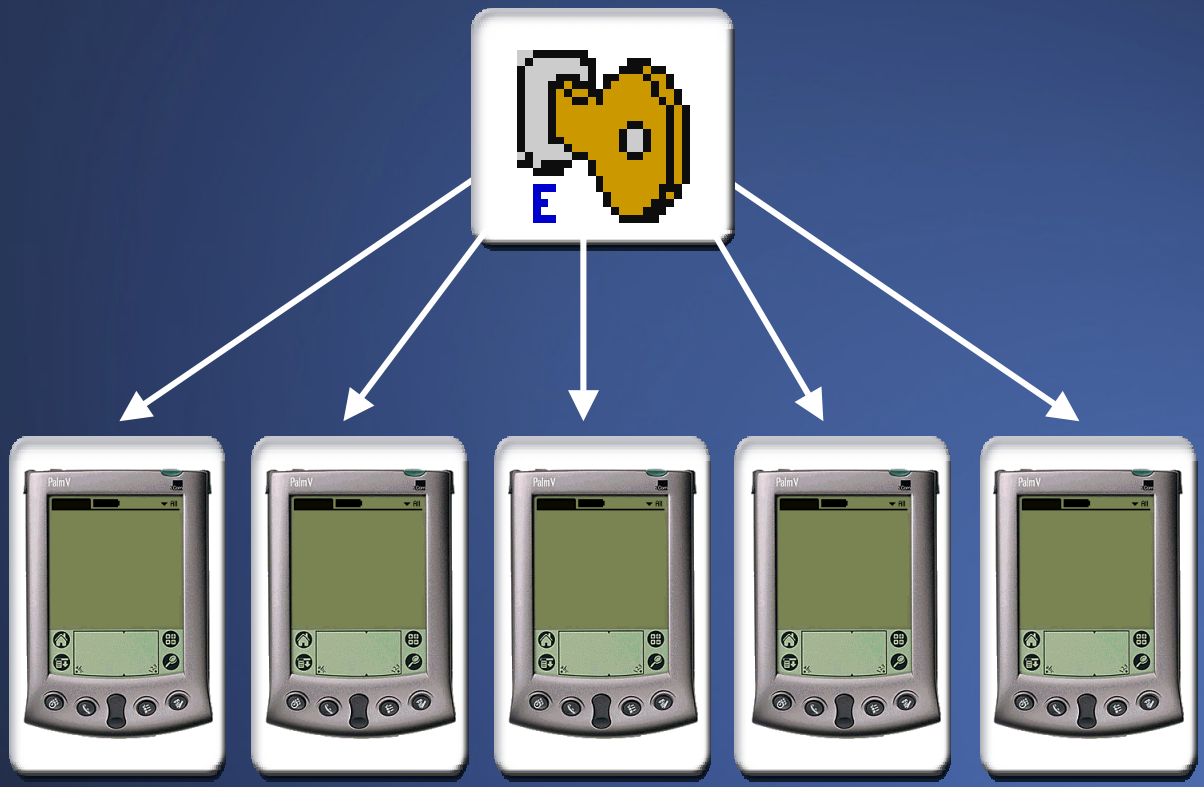
- If “Limit Applications” is checked then access is granted to a list of checked Apps for that category



Enterprise Access Controls Restrictor, Cory Krutis



- Pushing Down Security to Users via Syncronization



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Security Toolkits
Ntru, Steve Samson



- New choice for Palm security
  - Introducing NTRU Security Toolkit for Palm
- Faster and smaller than other solutions
  - Changes traditional speed vs. security tradeoffs
  - Core functions fit into ~16k of memory
- Easily implemented into applications
  - Commercial implementations completed in less than 1 week
- First Palm security toolkit to implement Rijndael Advanced Encryption Standard (AES)

Ntru Eliminates Key Storage

- Storing keys on a Palm is a significant risk
  - No matter how well hidden, they can be hacked
- NTRU eliminates the need to store keys
  - Can use a user-entered passphrase to generate a key
    - Not just unlock a stored key, as with other systems
  - Due to the unique mathematical properties underlying NTRU
- Significant new security improvement for many Palm applications
  - Especially where strong protection is a priority

Ntru Speed

- NTRU encryption is up to 40 times faster than other Palm cryptosystems
  - Eliminates waiting
  - Developers can maintain excellent user experience while delivering high security
- Enables developers to secure more data and authenticate more frequently
  - Security needs, not performance constraints, can drive choices
- Significantly enhances server scalability
  - Support more Palm users with fewer servers

Ntru Strong Security

- Eliminates traditional speed vs. security tradeoffs
- Based on more efficient mathematics than other systems, such as RSA and ECC
- Technology scrutinized by independent experts throughout the world
- Development team including leading ECC expert, and former RSA VP of Engineering
- Used by customers worldwide


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